

Evaluation of Wind Forecasts and Observation Impacts from Variational and Ensemble Data Assimilation for Wind Energy Applications

BRIAN C. ANCELL, ERIN KASHAWLIC,* AND JOHN L. SCHROEDER

Texas Tech University, Lubbock, Texas

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ABSTRACT

The U.S. Department of Energy Wind Forecast Improvement Project (WFIP) has recently been completed with the aim of 1) understanding the performance of different mesoscale data assimilation systems for lower-atmospheric wind prediction and 2) determining the observation impacts on wind forecasts within the different assimilation systems. Here an ensemble Kalman filter (EnKF) was tested against a three-dimensional variational data assimilation (3DVAR) technique. Forecasts lasting 24 hours were produced for a month-long period to determine the day-to-day performance of each system, as well as over 10 individual wind ramp cases. The observation impacts from surface mesonet and profiler/sodar wind observations aloft were also tested in each system for both the month-long run and the ramp forecasts.

It was found that EnKF forecasts verified over a domain including Texas and Oklahoma were better than those of 3DVAR for the month-long experiment throughout the forecast window, presumably from the use of flow-dependent covariances in the EnKF. The assimilation of mesonet data improved both EnKF and 3DVAR early forecasts, but sodar/profiler data showed a degradation (EnKF) or had no effect (3DVAR), with the degradation apparently resulting from a lower-atmospheric wind bias. For the wind ramp forecasts, ensemble averaging appears to overwhelm any improvements flow-dependent assimilation may have on ramp forecasts, leading to better 3DVAR ramp prediction. This suggests that best member techniques within the EnKF may be necessary for improved performance over 3DVAR for forecasts of sharp features such as wind ramps. Observation impacts from mesonet and profiler/sodar observations generally improved EnKF ramp forecasts, but either had little effect on or degraded 3DVAR forecasts.

1. Introduction

Accurate wind power prediction begins with high-quality atmospheric wind forecasts. The capability to produce accurate wind forecasts is thus crucially important to supporting cleaner energy practices and meeting the U.S. Department of Energy (DOE) goal of the nation providing 20% of its electricity from wind energy by the year 2030 (DOE 2008). Like any numerical weather prediction (NWP) problem, accurate wind forecasts primarily rely on two things: 1) the quality of the NWP model involved (e.g., resolution, physics) and 2) the accuracy of the initial and boundary conditions

from which the forecast evolves. Both of these aspects have recently been identified as key science priorities if meteorological support of the wind energy industry is to improve (Schreck et al. 2008; Shaw et al. 2009; Banta et al. 2013). A recently completed 2-yr study, the Wind Forecast Improvement Project (WFIP), was funded by the DOE with the aim of exploring the relative performance of wind forecasts associated with a number of modeling systems and various assimilated observation types. Here we present results regarding one aspect of the WFIP project that focused on the accuracy of forecast initial conditions, which subsequently depends on both the data assimilation technique used to ingest observations as well as the extent and quality of the observations themselves.

Two leading atmospheric data assimilation techniques that aim to achieve the most likely analysis with the minimum variance based on Bayesian statistics are three-dimensional variational data assimilation (3DVAR; Lorenc 1986) and an ensemble Kalman filter (EnKF; Evensen 1994). The primary difference between these

* Current affiliation: One Energy, LLC, Findlay, Ohio.

Corresponding author address: Brian C. Ansell, Texas Tech University, Department of Geosciences, Box 41053, Lubbock, TX 79409.
E-mail: brian.ansell@ttu.edu

schemes is that 3DVAR uses static climatological covariances to spread observational information spatially during assimilation, whereas the EnKF uses flow-dependent covariances calculated from an ensemble of forecasts. It has generally been found that within modern atmospheric modeling/data assimilation systems, flow dependence during assimilation can be advantageous. Both Whitaker et al. (2004) and Whitaker et al. (2009) were able to use ensemble data assimilation to effectively assimilate surface pressure data to reproduce the flow aloft using a coarse global model, which was found to be more effective than using 3DVAR. It was noted in both studies, however, that the benefits of flow-dependent covariances during assimilation are likely more substantial because the observations assimilated were sparse, a result first discussed in Hamill and Snyder (2000). At finer grid spacing (30 km), Meng and Zhang (2008a) and Meng and Zhang (2008b) showed improved forecast performance with an EnKF over that of 3DVAR with the Weather Research and Forecasting (WRF) Model for a mesoscale convective vortex and a month-long experiment verifying against radiosonde data. Meng and Zhang (2008b) assimilated only radiosonde data (relatively sparse), whereas Meng and Zhang (2008a) assimilated a variety of both surface and upper-air observational data of higher density.

Other studies using more simplified models have compared forecasts using ensemble data assimilation systems that use purely flow-dependent covariances to those that include “hybrid” combined flow-dependent/climatological covariances. Wang et al. (2007) and Hamill and Snyder (2000) both found that the hybrid technique becomes more accurate than the purely flow-dependent system as ensemble size decreases. Since prior research suggests the benefits of flow-dependent covariances within assimilation systems are reduced for both smaller ensemble size (and subsequent increased sampling error) and denser observations, it is crucial to understand what benefits an EnKF might retain over 3DVAR as the ensemble size within finer-scale operational systems are computationally constrained and relatively dense observations are available for assimilation. Here we study this behavior in the context of lower-atmospheric (~ 0 – 100 m above the surface) wind forecasts in support of improved wind power prediction. Further motivation for this examination is given by the fact that the finest resolution U.S. operational assimilation system [the Rapid Refresh/High Resolution Rapid Refresh (RR/HRRR); NOAA 2009] has recently evolved to incorporate flow-dependence in a hybrid ensemble-variational framework. Results here describing the strengths and weaknesses of ensemble and variational schemes relative to day-to-day wind and

ramp forecasts may help support further development of the RR/HRRR hybrid system. More generally, this evaluation could create a benchmark for testing hybrid mesoscale assimilation/forecasting systems as they become more standard and operationally feasible. Comparing 1–24-h wind forecasts from an EnKF to that of a 3DVAR system at relatively fine grid spacing (~ 10 km) is the first objective of this study.

The other aspect of any data assimilation system that affects the quality of the analysis is the number, location, and quality of the assimilated observations. It is generally assumed that more assimilated observations improve analyses, although a number of studies have developed both observation targeting and impact techniques that reveal localized regions where observations are expected to have the largest benefits. The formulation of “observation sensitivity” by Baker and Daley (2000) allowed one to diagnose within a 3DVAR system the impact of each assimilated observation, or various subsets of observations, on forecast error. This methodology was subsequently a basis for a number of observation impact studies to assess the most and least valuable observations in a 3DVAR system (Langland and Baker 2004; Tremolet 2008; Gelaro and Zhu 2009; Gelaro et al. 2010). A variety of studies have also revealed the importance of specific observations in ensemble-based data assimilation systems (Bishop et al. 2001; Ancell and Hakim 2007; Liu and Kalnay 2008). Here we aim to assess the aggregate effects on wind forecasts of two types of observational platforms: surface mesonet and upper-air profiler/sodar observations (a number of which were deployed specifically for the WFIP project). In lieu of using the observation impact techniques described above, here we use a data denial approach to assess the impacts of each type of observation. This removes one source of error involved with the assumption of linear perturbation evolution associated with the observation impact techniques described above. Evaluating the observation impacts of both mesonet and profiler/sodar data on 1–24-h wind forecasts within both an EnKF and a 3DVAR system is the second objective of this study. This paper is organized as follows: section 2 provides the methodology for the experiments performed, section 3 gives results and discussion, and the summary and conclusions are provided in section 4.

2. Methodology

a. The assimilation systems

This study utilizes both an EnKF and a 3DVAR data assimilation system. The EnKF is that of the Data

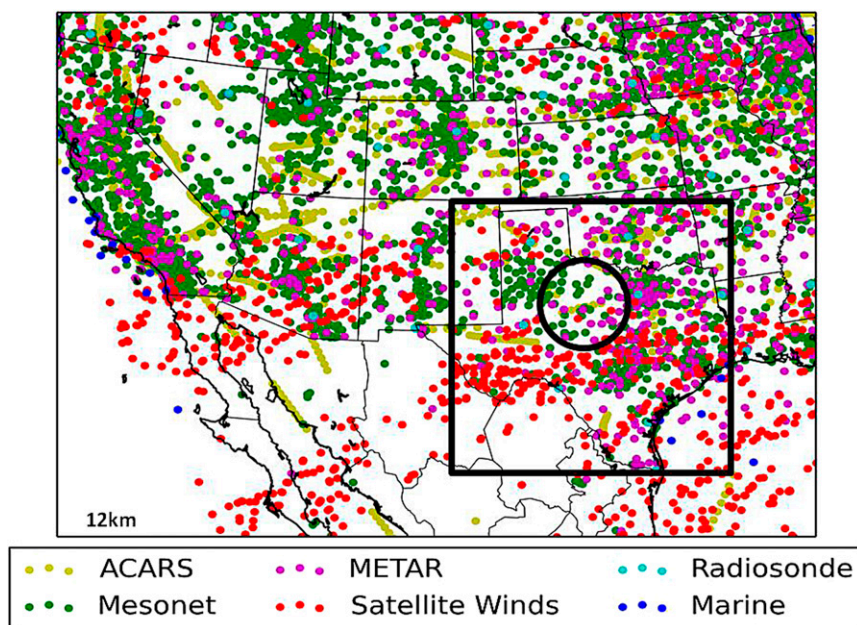


FIG. 1. The 12-km modeling domain used in this study. The black circle indicates the ramp verification area containing the 80-m meteorological towers, and the black box shows the verification area for the month-long experiments. A typical distribution of routine observations (valid at 0000 UTC 11 Dec 2013) representing those assimilated in this work is shown by the colored circles.

Assimilation Research Testbed (DART; Anderson et al. 2009). The ensemble adjustment Kalman filter (Anderson 2001) technique is used, which Anderson (2001) showed performs better than the traditional EnKF. To mitigate the effects of small sample size (Anderson and Anderson 1999), adaptive prior covariance inflation (Anderson 2009) in space and time are employed, as well as a Gaspari–Cohn localization radius (Gaspari and Cohn 1999) in both the horizontal and vertical direction (values of full width are used in this study). EnKF covariances are purely flow dependent and calculated from the ensemble. The ensemble size is 50 members, which is the same ensemble size within the real-time Texas Tech Ensemble Prediction System (TTEPS) that is able to run operationally on a large computing cluster. Every assimilation experiment in this study runs on a 6-h cycle. The ensemble is initialized through random perturbations drawn from climatological covariances within the WRF Model data assimilation system (WRFDA; Barker et al. 2012) added to the GFS analysis. Figure 1 shows the 12-km domain on which the EnKF system is run. In addition, there is a much larger parent 36-km domain (not shown, also initialized by adding perturbations from WRFDA climatological covariances to the GFS analysis) that stretches across the majority of the Pacific Ocean used to provide flow-dependent boundary conditions in the form of a

one-way nest to the ensemble members on the 12-km grid.

The 3DVAR system used in this work is that of the WRF-based global statistical interpolation system, version 2 (GSI; WRF Developmental Testbed Center 2010). Analogous to the EnKF, the GSI possesses a number of specific parameters that affect the horizontal and vertical extent of analysis increments (similar to EnKF localization), as well as a weight placed on the first-guess variance (similar to EnKF inflation). GSI covariances are static and are generated using the National Meteorological Center (NMC) method (Parrish and Derber 1992) with NAM 12-km model forecasts. These covariances were generated over many months and well represent the static covariance field in the Northern Hemisphere (WRF Developmental Testbed Center 2013, personal communication). Both data assimilation and forecasts within the GSI 3DVAR system occur over the same domain shown in Fig. 1. Boundary conditions to the GSI 3DVAR 12-km forecasts are provided by the 36-km EnKF mean forecast, insuring that the boundary conditions do not play a role in any differences of forecast skill between the 12-km EnKF and GSI forecasts.

Both the EnKF and GSI data assimilation parameters are tuned since the quality of analyses and forecasts in each system depend on such parameters and may vary

TABLE 1. Tuned assimilation parameters for both the DART EnKF and the GSI 3DVAR systems.

DART EnKF	
Horizontal localization	500 km
Vertical localization	5 km
Inflation: inf_initial	1.02
Inflation: inf_sd_initial	0.9
Inflation: inf_damping	0.9
Inflation: inf_lower_bound	1.0
Inflation: inf_jpper_bound	10 000 000
Inflation: inf_sd_lower_bound	0.9
GSI 3DVAR	
as_op	1.7, 1.1, 0.3, 0.9, 0.1
vs_op	1.7
hzscl_op	1.1, 1.1, 0.4

with different domain sizes, terrain complexity, etc. (the final tuned values for both the DART EnKF and GSI 3DVAR systems are shown in Table 1.) The tuning procedure was performed by cycling the assimilation system for a range of parameter values over 4 days of data assimilation cycles during the month-long verification period of December 2011 used in this study (16 assimilation cycles, all available surface and upper-air data was assimilated—see section 2c below for a description of observations). Extended 24-h forecasts were produced from each analysis, and the mean absolute errors measured against domain-wide METAR and radiosonde wind observations, averaged over all assimilation cycles, were inspected in order to choose the optimal parameters. The tuned parameters were those that produced the smallest errors over the forecast period. It should be noted that these parameters were tuned specifically for optimal wind forecasts, and different parameters might

exist that optimize other forecast aspects (although the same parameters found to optimize wind forecasts also optimized temperature forecasts).

The inflation parameters (six total) and both the vertical and horizontal localization radii were tuned for the EnKF. For GSI, there are three parameters (as_op, versus_op, and hzscl_op) that weight the first-guess variance and control vertical and horizontal localization. Unlike the EnKF for which a single parameter value collectively addresses all analysis variables, the GSI parameters exist independently for each the following analysis variables: streamfunction, velocity potential, temperature, surface pressure, and specific humidity. In turn, a very large number of runs were required to tune these parameter values. Figure 2 shows an example of two such tuning runs, one associated with the vertical localization within the EnKF, and the other associated with the horizontal localization (hzscl_op) within the GSI. These runs reveal the robust nature of the assimilation parameters in each system; as they are varied, the corresponding errors change the same way at each forecast hour. For the assimilation parameters more generally, values were varied until errors converged to a minimum, and these optimized values were used for all assimilation experiments. While a period of 4 days was used for the tuning runs here, extensive experience of the authors (Ansell et al. 2011, 2014) has shown that tuning over both longer and shorter time periods essentially provides the same optimal assimilation parameters for both the EnKF and GSI 3DVAR.

b. The forecast model

The modeling system used here is the Advanced Research WRF Model version 3.3 (Skamarock et al. 2008).

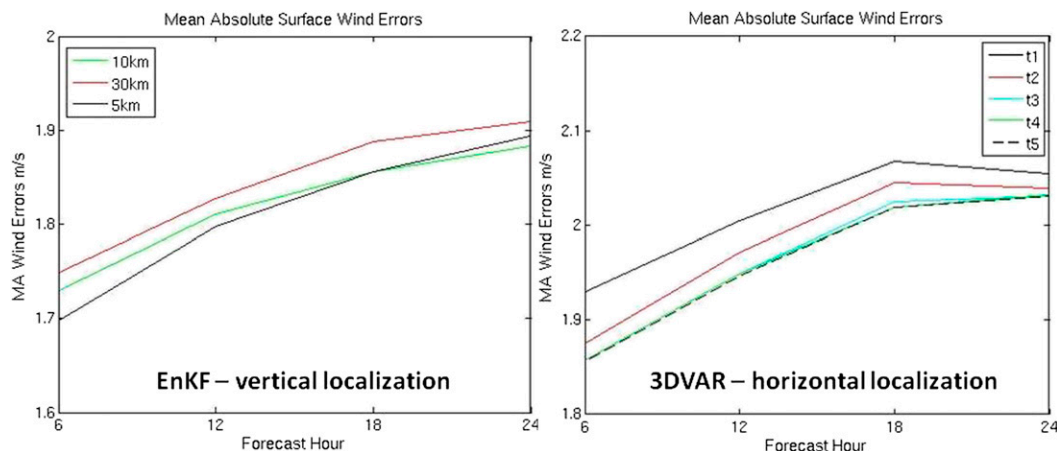


FIG. 2. Two examples of tuning runs for (left) DART EnKF vertical localization and (right) GSI 3DVAR horizontal localization performed over the first week of December 2011. The different tuning runs for GSI 3DVAR (t1–t5) correspond to different values of the secondary hzscl_op parameter, and are as follows for the full triple-valued parameter: t1(1.1, 0.2, 0.4), t2(1.1, 0.5, 0.4), t3(1.1, 0.8, 0.4), t4(1.1, 1.1, 0.4), and t5(1.1, 1.4, 0.4).

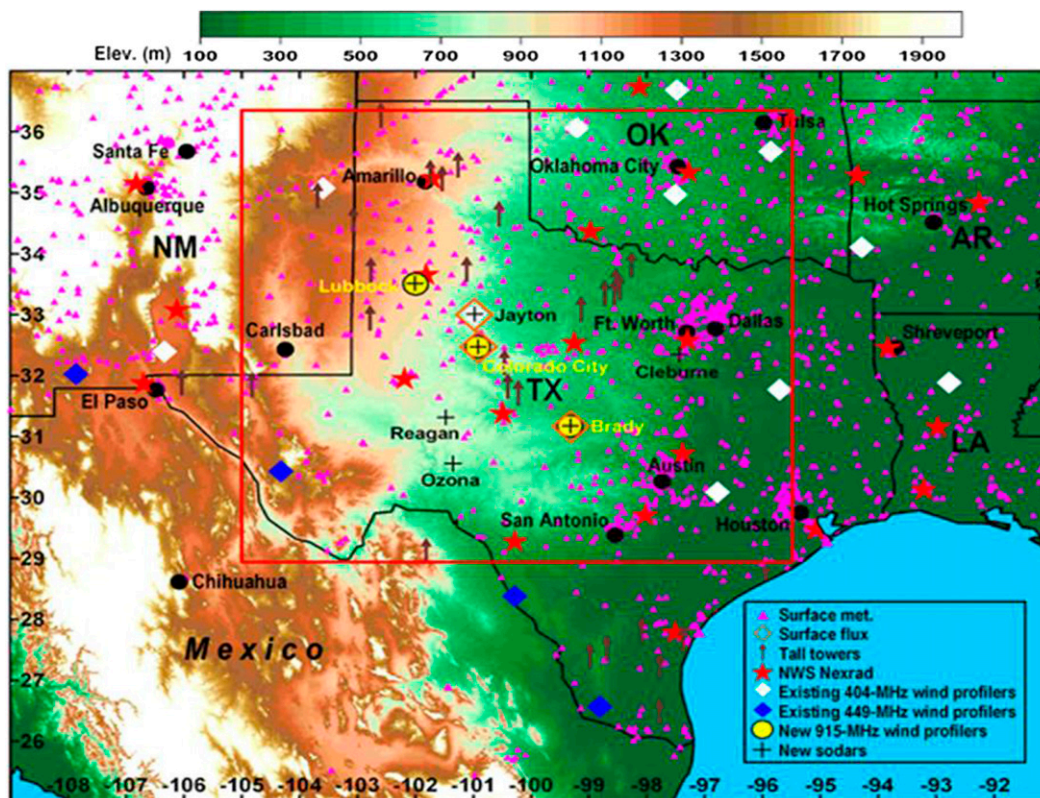


FIG. 3. The upper-air observations used within the WFIP project (including sodar and profiler observational platforms deployed specifically for WFIP).

The model physics used for all experiments are the Yonsei University (YSU) planetary boundary layer scheme (Hong et al. 2004), the Kain–Fritsch cumulus parameterization (Kain and Fritsch 1990, 1993), the Noah land surface model (Chen and Dudhia 2001), Thompson microphysics (Thompson et al. 2004), the Rapid Radiative Transfer Model (RRTM) longwave radiation scheme (Mlawer et al. 1997), and the Dudhia shortwave radiation scheme (Dudhia 1989). All model integrations use 38 vertical levels up to a model top of 50 hPa and are performed at 12-km grid spacing (with the exception of the runs performed on the 36-km EnKF domain that provide boundary conditions). Both domains possess a lowest model half-level existing about 20 m above the surface with roughly 16 levels in the lowest 1500 m, with vertical resolution ranging from about 150 m at that height to about 40 m near the ground. Ensemble boundary conditions for the outer 36-km domain are produced through perturbations about the GFS global model forecasts described in Torn et al. (2006).

c. The observations

The observations assimilated in this work are composed of a set of routine observations that are assimilated

in every experiment, as well as two sets of additional observations used in the observation impact experiments. The routine assimilated observations consist of METAR and marine temperature and wind data, satellite cloud-track winds, ACARS temperature and wind data, and radiosonde wind and temperature data. The two sets of additional observations consisted of 1) a mesoscale network of surface wind and temperature data, and 2) wind observations aloft from both profiler and sodar platforms. The sodar observations consist of wind speed and direction measured every 5–10 m from the surface to generally near 200 m, and the profiler observations (also wind speed and direction) exist at coarser resolution but to heights of around 5000 m (observations generally exist every 20–40 m up to about 3000 m, with sparser data about every 100 m from 3000 to 5000 m).

A typical distribution of observations is shown on the 12-km domain in Fig. 1 (with the exception of profiler and sodar observations). Figure 3 contains a plot from the WFIP project database showing both the existing profiler network and the additionally deployed sodar and profiler observations specific to the WFIP project (some additional data are also shown, as well as major

cities denoted by the black circles). Both the observations themselves, as well as the observation error variance values (obtained through the National Centers for Environmental Prediction), are identical between the EnKF and GSI 3DVAR experiments. Quality control of the observations was performed within the EnKF, which did not assimilate observations that 1) were different than the ensemble mean background field by more than three standard deviations, or 2) contained an elevation mismatch between the observation and model background terrain height field of 300 m or more. Bilinear interpolation in the horizontal and linear interpolation based on pressure in the vertical was used as the forward operator to estimate the model background field in each system at the observation locations.

d. Experimental setup

A variety of experimental runs are performed in order to compare the performance of the DART EnKF and GSI 3DVAR, as well as to test the observational impacts of both mesonet surface data and profiler/sodar observations in each system. Two sets of experiments are performed: 1) a month-long continuously cycled integration during December 2011 for all cycles from 0000 UTC 2 December to 1200 UTC 31 December (119 assimilation cycles), and 2) individual periods of cycled assimilation runs that capture 10 individual wind ramp events (7 up ramps and 3 down ramps). These ramp events were selected by WFIP private industry project partners as the 10 highest priority wind ramps based on their internal ramp thresholds over the winter 2011/spring 2012 season. These thresholds were based on observed power (mW) increases over 15-, 60-, and 180-min intervals, and classified as small, medium, and large as determined by a range of values set by the industry partners based on professional experience. Each of these wind ramps were synoptically forced, either through frontal/dryline passages or synoptic-scale forcing of the surface pressure field, and do not include ramp cases from other phenomena such as convectively-driven outflow events. The 24-h forecasts were run from all analyses for both the month-long integration as well as each individual cycling period in order to evaluate any dependence of performance on forecast lead time. Furthermore, a spinup period of 2 days was run after ensemble initialization on a 6-h assimilation cycle in both the EnKF and 3DVAR systems for both the month-long and wind ramp experiments, mostly to allow flow dependence to build within the EnKF runs.

In this study, forecast comparisons are made in the 1–24-h forecast range. Analysis errors are not shown for two reasons: 1) because the analysis fit to observations is controlled through the data assimilation parameters,

making a fair comparison of analysis quality difficult among the two systems, and 2) GSI 3DVAR does not update diagnostic fields (the WRF surface-based variables used to calculate error) during assimilation, making a direct surface analysis comparison between the systems impossible. While analysis quality could be judged through independent observations [as performed in Ancell et al. (2011)], here we assimilate all data toward measuring the operational skill of the systems, and thus focus on comparing forecast quality. Nonetheless, we have thoroughly checked the fit to observations for both systems (using the first model time step with updated diagnostic fields for the GSI runs) both at the surface and aloft for the background and analysis. Both the GSI and EnKF experiments exhibit a substantially closer fit to observations at analysis time, indicating the appropriateness and correctness of both assimilation systems.

The month-long runs are verified through mean absolute errors (calculated through u and v components) against wind observations from surface METAR stations (scattered throughout the black box in Fig. 1). The wind ramp events are verified against wind observations from twenty 80-m meteorological towers located on existing wind farms (contained within the black circle in Fig. 1). Four aspects of the wind ramps are used to measure the forecast skill of each experimental run: 1) ramp onset, 2) ramp duration, 3) ramp magnitude, and 4) ramp maximum wind speed. This object-oriented verification approach was taken because large mean absolute errors were found with wind ramps that possessed relatively small timing errors (but showed otherwise well-forecast characteristics such as ramp duration and magnitude) compared with other runs that appeared to capture the overall ramp much less accurately. Furthermore, this verification strategy has the potential to allow wind farm operators to decide which forecasts are best for their purposes based on the skill associated with the different ramp aspects. The various metrics were calculated with respect to each tower individually.

3. Results and discussion

a. Month-long period of December 2011

The mean absolute 1–24-h wind errors measured against METAR surface observations for the EnKF and 3DVAR month-long experiments, averaged over the 119 assimilation cycles, are shown in Fig. 4. The EnKF performs better than 3DVAR averaged over the entire forecast window (1.48 vs 1.55 m s^{-1} , statistically significant at the 95% confidence level using a one-sided

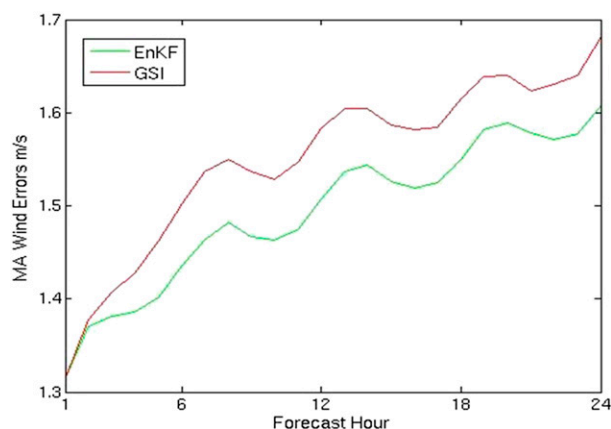


FIG. 4. The mean absolute 1–24-h wind errors for the EnKF (green line) and 3DVAR (red line) month-long routine observation experiments averaged over 119 assimilation cycles in December 2011.

Student's *t* test). Errors from both systems are about the same for the first 2 h of the forecast window, after which point the EnKF begins to show lower errors and increasing improvement over 3DVAR until the roughly 6-h forecast time. A roughly constant improvement is then achieved by the EnKF within the 6–24-h time frame. Since the same boundary conditions are used for both systems, the near steady advantage within the 6–24-h window suggests the benefits achieved through the EnKF during data assimilation actually increase during that time in order to offset the decreasing benefit one might expect from similar boundary conditions (as seen in Fig. 2).

The improved performance at the majority of forecast lead times of the EnKF is likely caused by the flow-dependent covariances it uses during assimilation. Figure 5 shows examples of the background field and the analysis increments of both 500-hPa geopotential height (GPH; 0600 UTC 23 December) and 80-m meridional wind component (1800 UTC 3 December) for both systems. At 500 hPa the analysis increments clearly show an adjustment made to the character of the trough over the U.S. Southwest in both systems, and appear to have similar magnitudes and spatial scales. Within both this example and other initializations (not shown), it is difficult to subjectively ascertain any flow-dependent advantages the EnKF provides; analysis increments are generally similar in terms of location, size, and magnitude. This likely indicates the ability of 3DVAR to perform relatively well compared to the EnKF aloft where largely balanced flow results in covariance relationships that are somewhat time independent. The observation density and localization parameters play a role here as well; observations are dense enough so that EnKF analysis increments do not obviously capture flow

features that could not possibly be identified through climatological covariances (e.g., a dipole surrounding the base of a trough achieved through the assimilation of a single observation).

The effects of flow dependence within the EnKF are substantially clearer near the surface, however. In the example shown in Fig. 5 at 1800 UTC 3 December, a strong north–south pressure gradient exists just north of the Texas Panhandle, and is associated with a southward-moving arctic cold front adjacent to the high terrain to the west. This common pattern that impacts the U.S. southern plains is described in Colle and Mass (1995), and is typically associated with a sharp wind shift from westerly to northerly and sometimes a substantial wind ramp in the west Texas area. As expected, the EnKF flow-dependent 80-m meridional wind analysis increment exhibits a feature that is strongly aligned along the southward surging cold front, possessing values of over -7 m s^{-1} . The negative values indicate that the EnKF assimilation procedure significantly strengthened the northerly flow along and behind the frontal position within the first-guess field. The 3DVAR analysis increment is in a similar location but is clearly more isotropic in nature and less sharply aligned with the front at analysis time. Note the similar sharpness in the EnKF relative to 3DVAR along the Pacific cold front along the wind shift in central Texas at the same time. It follows that the EnKF analysis increments are likely more realistic, leading to an improved evolution in time relative to that of 3DVAR, particularly during early forecast hours.

To understand whether the relative skill of surface wind forecasts in the EnKF and 3DVAR is replicated aloft, Fig. 6 shows the 6–24-h 500-hPa mean absolute errors for temperature, geopotential height, and winds averaged over the month-long verification period. The relative behavior of errors between the systems is similar for all three variables: the EnKF shows an advantage at 6-h forecast time, but this advantage disappears and becomes a negligible difference at 12 h and for the rest of the forecast window (although for wind the EnKF advantage lasts through 12 h). Interestingly, this reveals that although the governing synoptic flow is no better in either system in the second half of the 24-h forecast window, the surface wind field remains significantly better within the EnKF throughout that portion of the period. Based on the above discussion regarding the more flow-dependent nature of EnKF analysis increments near the surface, this result shows that the benefits of such flow dependence linger well into the forecast period (at least through 24 h), even after the skill of the EnKF and 3DVAR systems aloft become the same. This suggests the biggest benefits of ensemble,

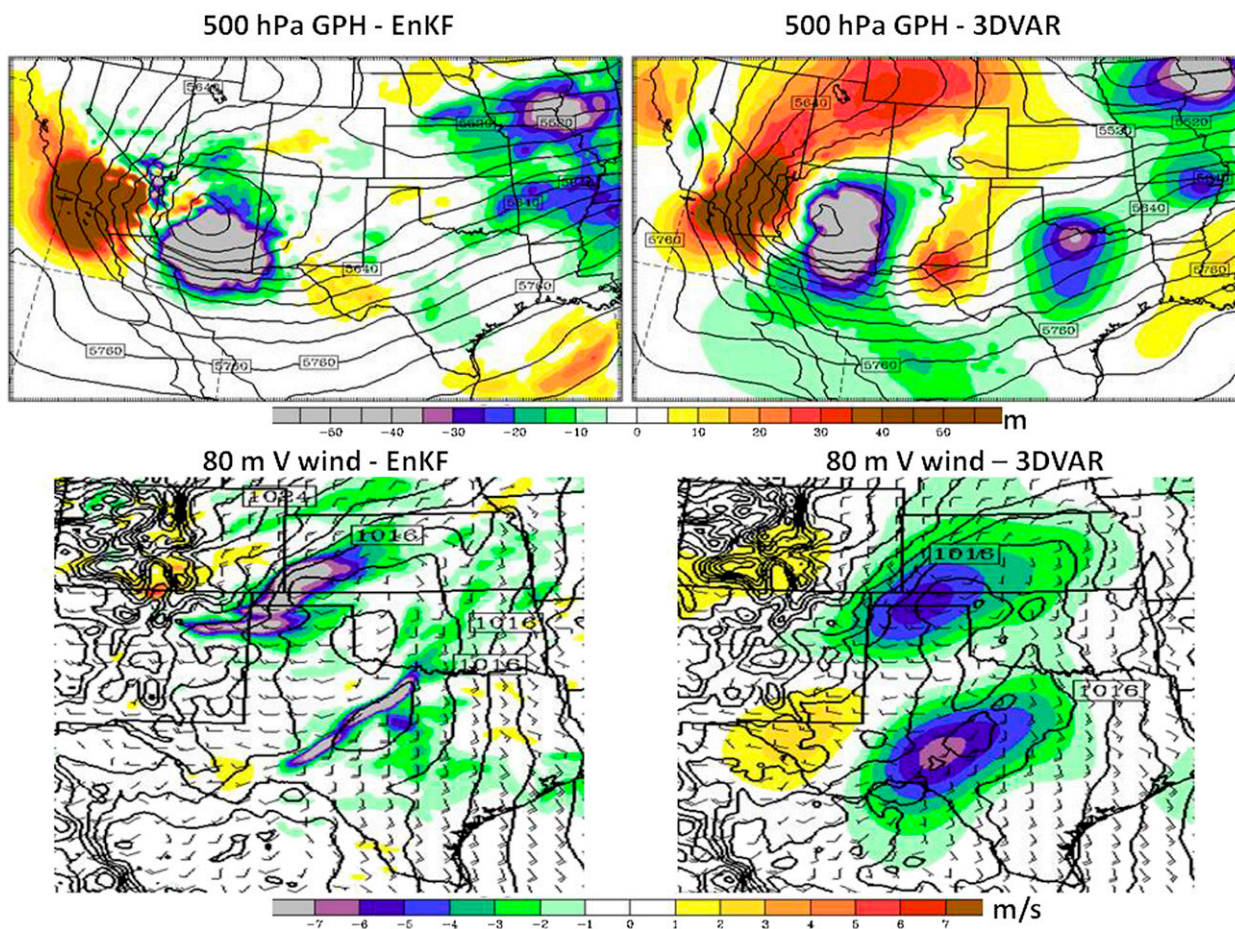


FIG. 5. (top) The 500-hPa background field (black contours, contour interval is 30 m) and analysis increments (shaded) valid at 0600 UTC 23 Dec and (bottom) the sea level pressure background field (black contours, contour interval is 2 hPa), background 80-m wind barbs, and 80-m meridional (V wind) analysis increments (shaded) valid at 1800 UTC 3 Dec for the EnKF and 3DVAR analyses.

flow-dependent data assimilation for lower-atmospheric wind prediction, at least through a day's time, is coming from the assimilation of surface observations. One interesting feature of note is the decrease of error within the GSI runs from 6 to 12 h. While errors do not decrease

in the EnKF runs over that same period, they do grow slower over that time than they do from 12 to 24 h. We speculate this behavior is related to balance adjustment inherent to any statistical assimilation procedure; after roughly a 6-h period of geostrophic balance adjustment

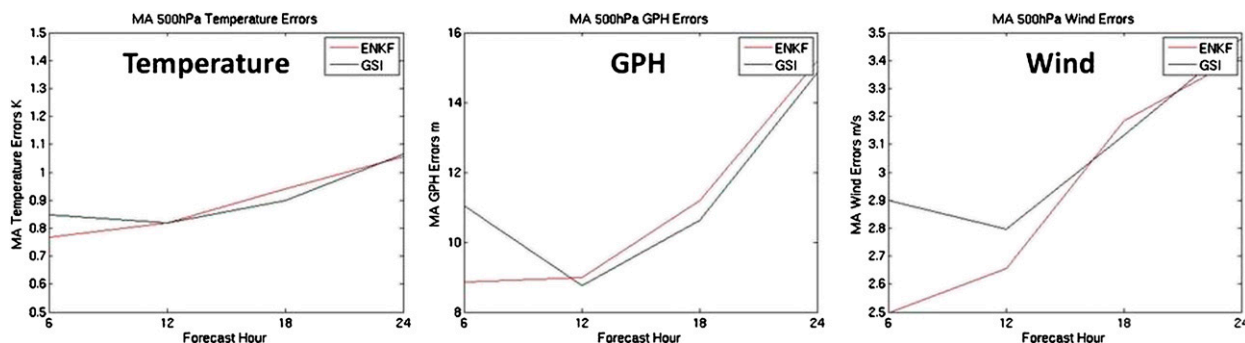


FIG. 6. The mean absolute 6–24-h temperature, geopotential height (GPH), and wind errors for the EnKF (red line) and 3DVAR (black line) month-long routine observation experiments averaged over 119 assimilation cycles in December 2011.

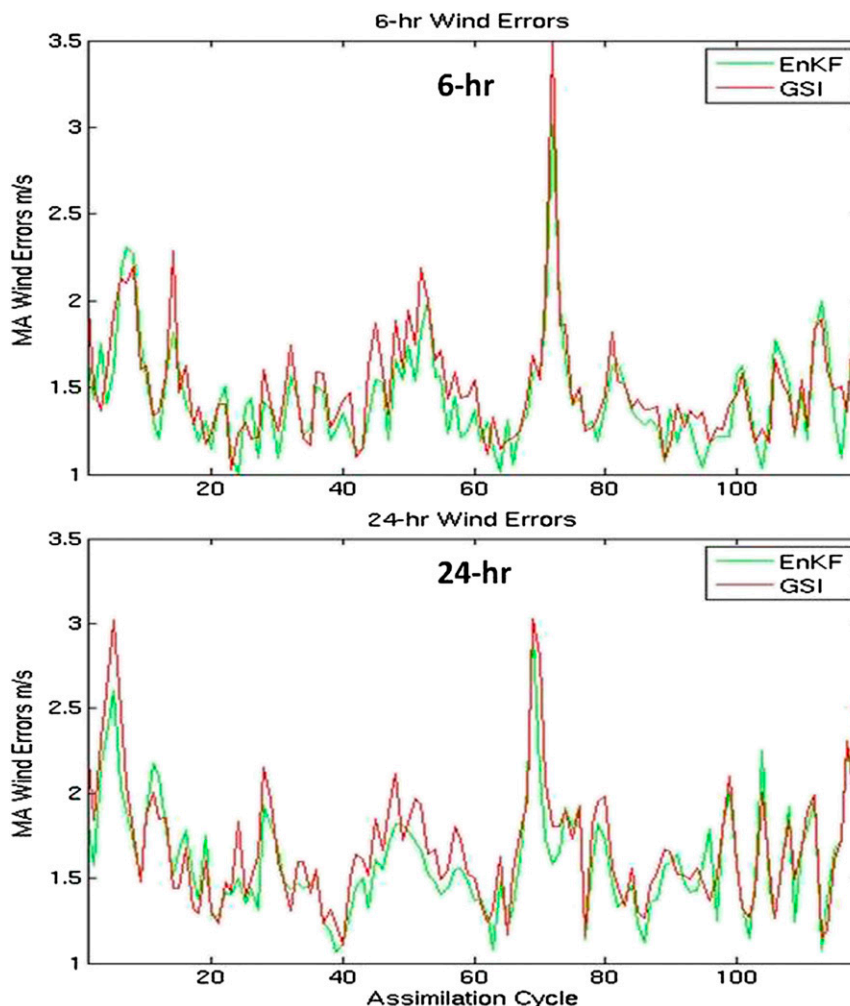


FIG. 7. The mean absolute EnKF (green line) and 3DVAR (red line) wind errors for all 119 assimilation cycles in December 2011 at both 6- and 24-h forecast time for the month-long routine observation experiments.

with errors growing somewhat rapidly (the analysis fit to observations was significantly better for both systems), the adjustment toward the actual balanced state seems to be achieved sometime within the 6–12-h window. That GSI errors actually decrease over that span suggests the adjustment process is more extreme than within the EnKF system, which shows errors that still increase but at a slower rate.

Although the EnKF exhibits better performance at nearly every forecast hour, neither system consistently performs better at a single forecast time. Figure 7 shows the mean absolute errors averaged over the verification area for all 119 assimilation cycles at both the 6- and 24-h forecast time. Note that both forecast times show cycles for which the EnKF was better than 3DVAR, and vice versa. This provides a good next step to further research into the benefits of each system for surface wind

forecasts. Since the different systems perform better at different analysis times, it is possible the success of each system depends on the temporal variation of the flow, and this will be further evaluated in the future. Further motivation for this evaluation is warranted since synoptic-scale flow-dependent predictability was clearly demonstrated in Ancell and McMurdie (2013), and may be manifesting itself here within the different data assimilation systems.

Figure 8 depicts the mean absolute errors for both the EnKF and 3DVAR for the month-long control run (no mesonet or profiler/sodar observations, only routine observations) as well as the runs that assimilate mesonet observations and profiler/sodar observations (with separate experiments) in addition to the routine observations. Both systems show the same result with regard to mesonet data (i.e., early forecasts are improved, but

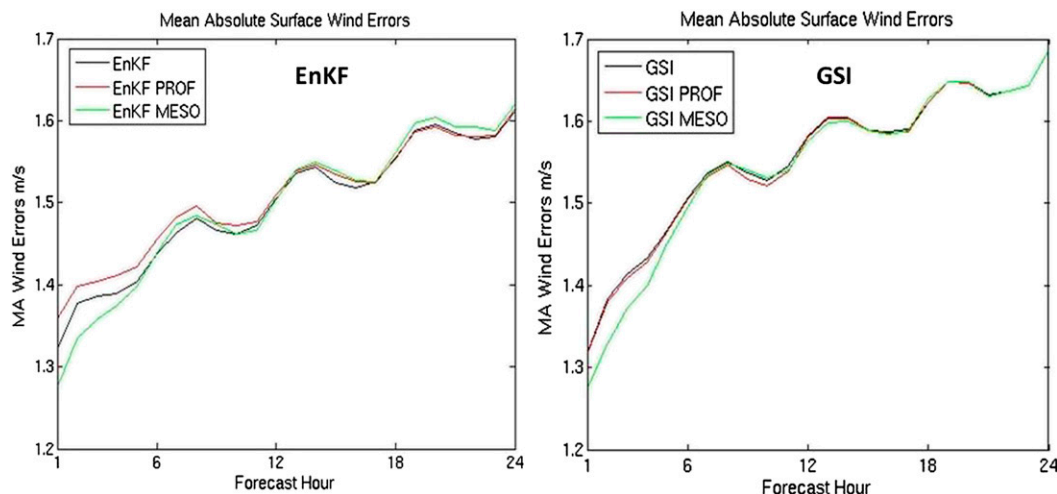


FIG. 8. The mean absolute 1–24-h wind errors for the month-long control run (black line, routine observations only), the run that assimilates mesonet observations in addition to routine observations (green line), and the run that assimilates profiler/sodar observations in addition to the routine observations (red line) for both EnKF and 3DVAR.

there is virtually no effect beyond 6 h). The improvements in both systems prior to 6 h maximize at the 2-h forecast time (2.9% improvement in EnKF, 3.6% in 3DVAR, statistically significant at the 90% confidence level). These small improvements perhaps reflect the findings of Knopfmeier and Stensrud (2013), who showed limited analysis benefits from assimilating mesonet data since the observation density was large relative to the scale of analysis increments. Here it seems such limited success of mesonet assimilation found in that study extends into early forecast time, and disappears altogether in an average sense with regard to surface wind forecast skill roughly 6 h into the forecast window. The early forecast improvements that do occur will have further reduced benefits to the wind power forecasting industry since the time to produce such early forecasts even with modern computational resources is usually comparable the clock time up to that point.

Results with regard to the assimilation of profiler/sodar data were mixed. The 3DVAR runs showed practically no impact at any time while the EnKF runs exhibited a small degradation through the first 12 h of the forecast. Since more assimilated observations generally improve forecasts, it is somewhat unexpected that profiler/sodar observations degrade early forecasts in the EnKF. Based on results from Ancell (2012) who showed that bias removal at assimilation time can degrade forecasts, we hypothesize the presence of bias in the current runs causes the observed early forecast degradation. Table 2 shows the background mean error, or bias, calculated against both METAR surface and 80-m tower wind observations for the month-long experiment

for the routine EnKF runs as well as the runs that assimilated profiler/sodar observations. For both 10- and 80-m winds, very similar wind speed biases are produced in the background field for each experiment, indicating the systematic error to which all runs are subject. However, biases are more substantial at 80 m, ranging from magnitudes of about $2\text{--}5\text{ m s}^{-1}$. The sign of the biases indicate winds are too slow at all times except for 1800 UTC, when winds are too fast. At the surface, a fast bias reaching magnitudes of almost 1 m s^{-1} exists for the 0000, 0600, and 1200 UTC cycles, while there is practically no bias at 1800 UTC. These biases likely point to issues with the diurnal timing of momentum flux involved with the boundary layer parameterization. Furthermore, analysis bias at 80 m is actually made worse compared to the background bias (not shown) for the experiments without the assimilation of profiler/sodar data, demonstrating how the removal of the surface fast bias during assimilation is spread aloft through ensemble covariances.

TABLE 2. Mean background (6-h forecast) errors calculated against both METAR and 80-m tower observations averaged over the 0000, 0600, 1200, and 1800 UTC assimilation cycles for both the routine and profiler/sodar experiments.

	Assimilation cycles (UTC)			
	0000	0600	1200	1800
METAR 10 m				
Routine	0.87	0.90	0.74	−0.06
Profiler/sodar	0.90	0.95	0.79	−0.01
Tower 80 m				
Routine	−2.06	−5.30	−2.93	1.90
Profiler/sodar	−2.11	−4.97	−2.58	2.00

Ancell (2012) showed that bias removal prior to assimilation produced about a 2%–4% (depending on resolution) increase in RMS surface wind forecast error that appeared by the first forecast hour, a degradation that is similar to that seen here in Fig. 8. That study hypothesized the adjustment back to the model attractor after bias removal resulted in the degradation, which can occur away from assimilated observation locations since bias is spread during assimilation. Since relatively large background bias has been demonstrated here in the boundary layer, we speculate the same issue is occurring when assimilating boundary layer winds (with profilers and sodars) relative to the runs without the ingestion of such data. Interestingly, the degradation due to the assimilation of profiler/sodar observations disappears for the 1800 UTC cycles (not shown; the degradations for the 0000, 0600, and 1200 UTC cycles strongly resembles that of Fig. 8). The fact that the single time for which assimilating profiler/sodar data shows no forecast degradation is also the one time no surface bias exists supports the notion that bias removal during assimilation might be detrimental. For every other time, the surface bias is opposite in sign to that at 80 m, resulting in opposing tendencies for bias removal at both levels when both surface and profiler/sodar data is assimilated. The lack of surface bias at 1800 UTC allows bias removal aloft to occur unopposed during assimilation, producing the forecast degradation. Interestingly, no degradation occurs with the 3DVAR runs at any time, suggesting that the use of climatological covariances does not produce the degradations involved with bias removal. Thus, flow dependence is actually detrimental in this case when relatively large bias is involved. Results from Ancell et al. (2014) support this idea, as they also showed how representative errors in complex terrain were exacerbated when flow-dependent covariances were used (instead of climatological covariances). It seems that the advantages of flow dependence during assimilation can become disadvantageous in the presence of factors that are known to degrade forecasts (e.g., bias or representativeness error).

These results underscore a critical need to address lower-atmospheric model wind bias during assimilation if profiler/sodar observations are to provide value to wind forecasts. It should be noted that Hu et al. (2013) describe updates to the YSU boundary layer scheme implemented in later versions of WRF than that used in this study that provide better nocturnal lower-atmospheric wind forecasts. However, rerunning the experiments with WRF, version 3.5.1 [a version of WRF that incorporates the upgrades described in Hu et al. (2013)] produced almost no change in the biases observed here. It is possible that other boundary layer

parameterizations may produce smaller near-surface wind biases since ultimately errors in the boundary layer physics are the cause for the biases in the first place. In any case, the degradation of forecasts through the removal of bias suggests improvements to boundary layer parameterizations are fundamentally important to fully realizing the benefits of boundary layer winds through ensemble data assimilation.

b. Wind ramp forecasts

Forecasts from the EnKF and 3DVAR were compared for 10 wind ramp cases, and mesonet and profiler/sodar observation impacts were evaluated in each system for these same ramp cases. Figure 9 shows an example of the typical differences between the two systems for one such wind ramp forecast initialized at 0000 UTC 3 December 2011 valid at a single 80-m meteorological tower in the ramp verification area shown in Fig. 1. A very large up ramp began around 4 h into the forecast period with the approach of a Pacific cold front, with winds increasing for roughly 5 h before leveling off at about 45 mph (20.1 m s^{-1}) for nearly 3 h. A large and rapid down ramp immediately followed the cold frontal passage (marked by the wind shift from southerly to westerly at about 12-h forecast time in Fig. 9b), with a subsequent 6-h plateau of roughly 20-mph (8.9 m s^{-1}) winds (with perhaps a slight increase during the middle of this plateau) before a slow decrease in wind speed over the last few hours of the forecast window.

Both the EnKF and 3DVAR runs captured the timing of the up ramp very well, and the EnKF seemed to better approximate the 3-h period of maximum winds (although both systems substantially underestimated the highest wind speeds). The 3DVAR run, however, better captured the slight increase of winds following the sharp down ramp, with the EnKF showing only a smooth and sustained down ramp after the period of maximum winds. It is likely that the smooth characteristics of the EnKF forecast (demonstrated most profoundly in the plot of wind direction in Fig. 9b) result from ensemble averaging, perhaps revealing unrealistic behavior but potentially statistically outperforming other deterministic runs such as that of the 3DVAR experiment [discussed in Ancell (2013)]. Contrary to the EnKF, both the 3DVAR run and the observations indicate sharp wind shifts associated with the frontal passage that caused the ramps. Here we composite a number of wind ramp cases to understand which system better predicts various ramp attributes, and whether mesonet or profiler/sodar data produces any impact on the ramp forecasts in each system.

Figure 10 shows the performance of the EnKF versus 3DVAR, as well as the different observation impacts in

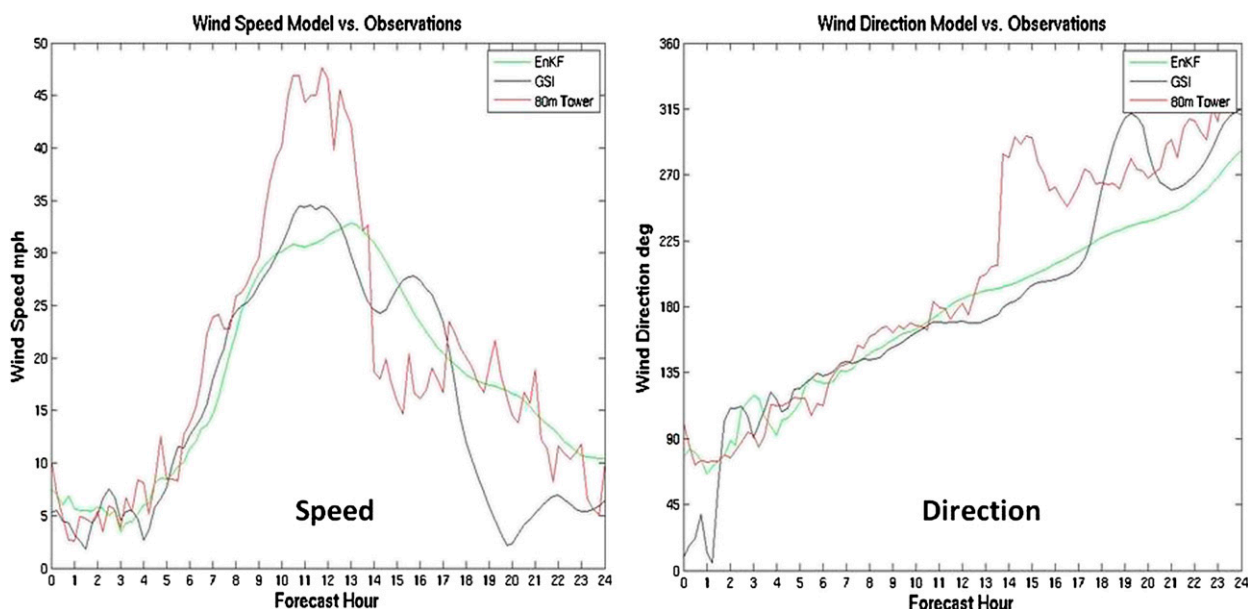


FIG. 9. Example wind ramp speed and direction forecasts from the EnKF (green line) and 3DVAR (black line) systems initialized at 0000 UTC 3 Dec 2011 valid at the location of a single 80-m meteorological tower. The observed wind speed and direction from the tower (red lines) are also shown.

each system, for absolute ramp onset (h) and duration (h) errors. Figure 11 shows the same comparisons for absolute ramp magnitude (m s^{-1}) and maximum wind speed (m s^{-1}) errors. The results are composited by three categories of forecast lead time (early: 0–9 h, middle: 9–15 h, and late: 15–24 h), and data were placed into each of these forecast lead time bins based on the forecast hours over which the majority of the observed ramp occurred. Results averaged over all lead times are also presented in the same figures (indicated by the colored circles in each figure). Data from all twenty 80-m towers were used for this composite verification, although there were some cases for which some of the towers had missing data. For this forecast wind ramp analysis, since statistical significance is very challenging to appropriately apply to a sample size of 10, we have chosen a rough set of guidelines to reveal the relative performance of the various experimental runs from the composite of cases used here. Onset and duration errors are viewed as “significant” if they exceed 15 min, whereas ramp magnitude and maximum wind speed errors are viewed as significant if they exceed 0.5 m s^{-1} . These guidelines are also influenced by what we perceive would be important with regard to error to wind farm operators. It should be noted that without formal statistical testing, however, these results are only suggestive of the forecast performance of the various runs more generally, and are only meant to reveal any clear advantages the different experiments may possess.

The 3DVAR runs appear to clearly outperform the EnKF for both ramp onset and duration (Fig. 10). A roughly 30-min advantage for early lead times increases to a 45-min advantage for middle and late lead times for both attributes (with an overall 30–45-min improvement). Similar 3DVAR improvements are shown in Fig. 11 for ramp magnitude and maximum winds, with small errors at early lead times growing to over 0.5 m s^{-1} averaged over the 10 cases. We speculate this EnKF degradation arises from the use of the ensemble mean in the presence of sharp features (the wind ramps), an effect shown to be detrimental to forecasts in Ancell (2013). In that study, ensemble mean surface wind speed errors of over 10 m s^{-1} were found in areas of high winds associated with midlatitude cyclones, a result of ensemble averaging in the presence of growing nonlinearity in the evolution of ensemble perturbations. Such nonlinearity, and associated ensemble mean errors, were shown to increase smoothly over the 24-h period, further supporting the role of ensemble averaging in the degradation of the EnKF here since 3DVAR improvements are also seen to increase smoothly in Figs. 10 and 11. While we suspect flow-dependent data assimilation is superior more generally based on Fig. 4 and numerous other studies, it appears the ensemble mean degrades forecasts with regard to spatially and temporally less common sharp features such as wind ramps, particularly with regard to specific feature attributes (e.g., ramp onset or duration).

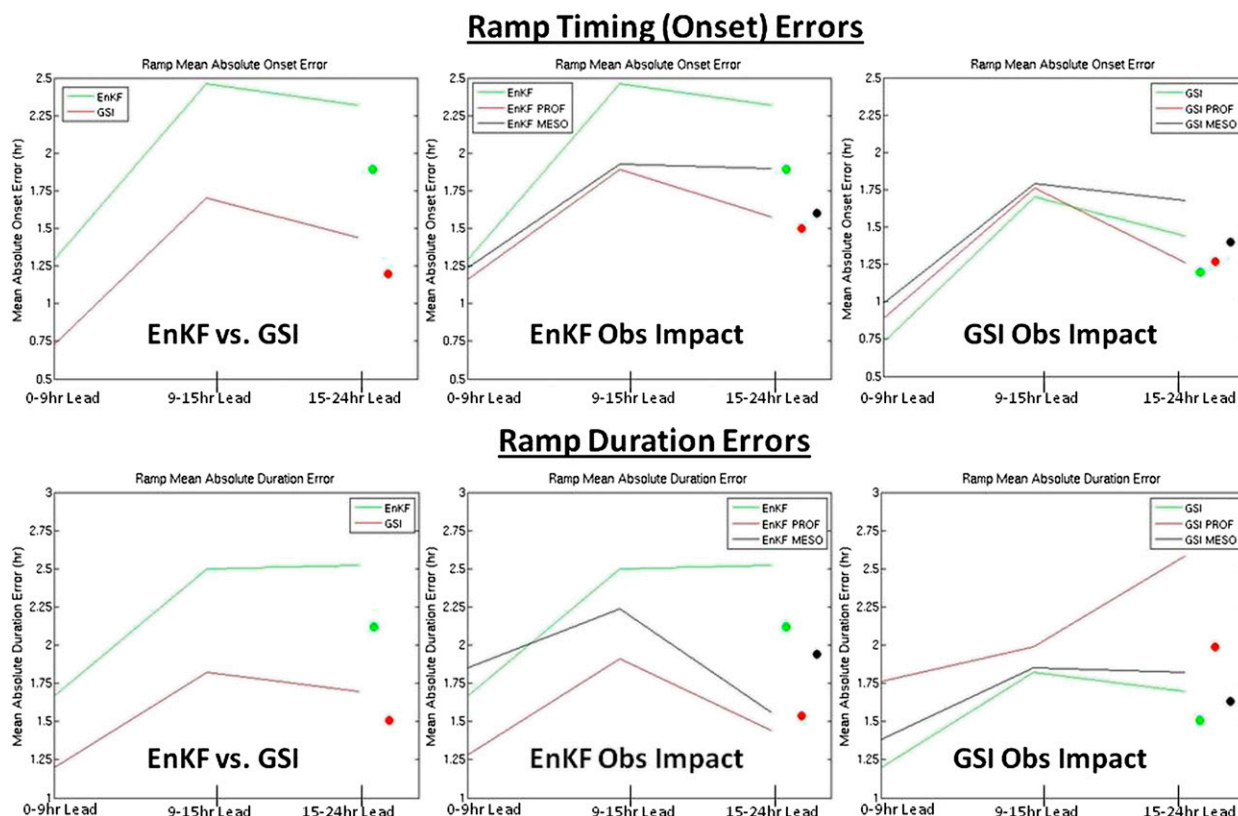


FIG. 10. Ramp timing (onset) and duration errors averaged over all wind ramp cases for the EnKF and 3DVAR routine and observation impact experiments. The colored circles represent the average error over the three lead-time categories shown (0–9, 9–15, and 15–24 h).

That ensemble averaging potentially degrades wind ramp forecasts suggests that best member techniques might improve EnKF forecasts [a technique shown to be successful in [Ansell \(2013\)](#)]. Determining such best members is a nontrivial problem; members can be selected based on aspects of the wind ramp itself, related larger-scale features of the flow, or even model state variables over the entire domain. If the potential of the EnKF for accurate wind ramp forecasts is to be realized, this is large and an important extension of this study, one we plan on exploring in the near future. We aim to evaluate members chosen based on their closeness to the mean [as done in [Ansell \(2013\)](#) for midlatitude cyclones] as well as those chosen based on the magnitudes of their errors in sensitive regions determined through the use of ensemble sensitivity analysis ([Ansell and Hakim 2007](#)).

The wind ramp observation impact experiments produce different results for the EnKF and 3DVAR. Within the EnKF, improvements are found at most lead times for both types of additional observations (profiler/sodar and mesonet). These improvements are most pronounced at middle and late lead times for ramp onset and duration, where 30–60-min advantages are seen, with the largest benefits resulting from the assimilation

of profiler/sodar data. For ramp magnitude and maximum winds, improvements are largest at early lead time, mostly for the assimilation of mesonet data (showing a 0–5–1.0 m s^{-1} improvement). In any case, observation impacts were mostly positive and never increased error by a substantial margin. The opposite is true for the 3DVAR observation experiments. For all four ramp attributes, differences in errors were for the most part negligible or slightly worse. The most obvious difference was the 30–60-min degradation at early and late lead times of ramp duration error when assimilating profiler data. In any case, unlike the EnKF experiments, the 3DVAR forecasts essentially showed no improvement when assimilating profiler/sodar and mesonet data. This further indicates the relative potential of EnKF with its positive observation impacts, as 3DVAR appears to be “saturated” with regard to whether additional assimilated observations improve wind ramp forecasts. In both experiments, it is quite interesting how different observational impacts are seen with regard to different ramp attributes since it might be expected that such attributes be strongly correlated for each event. One method to examine this behavior is through ensemble sensitivity analysis of

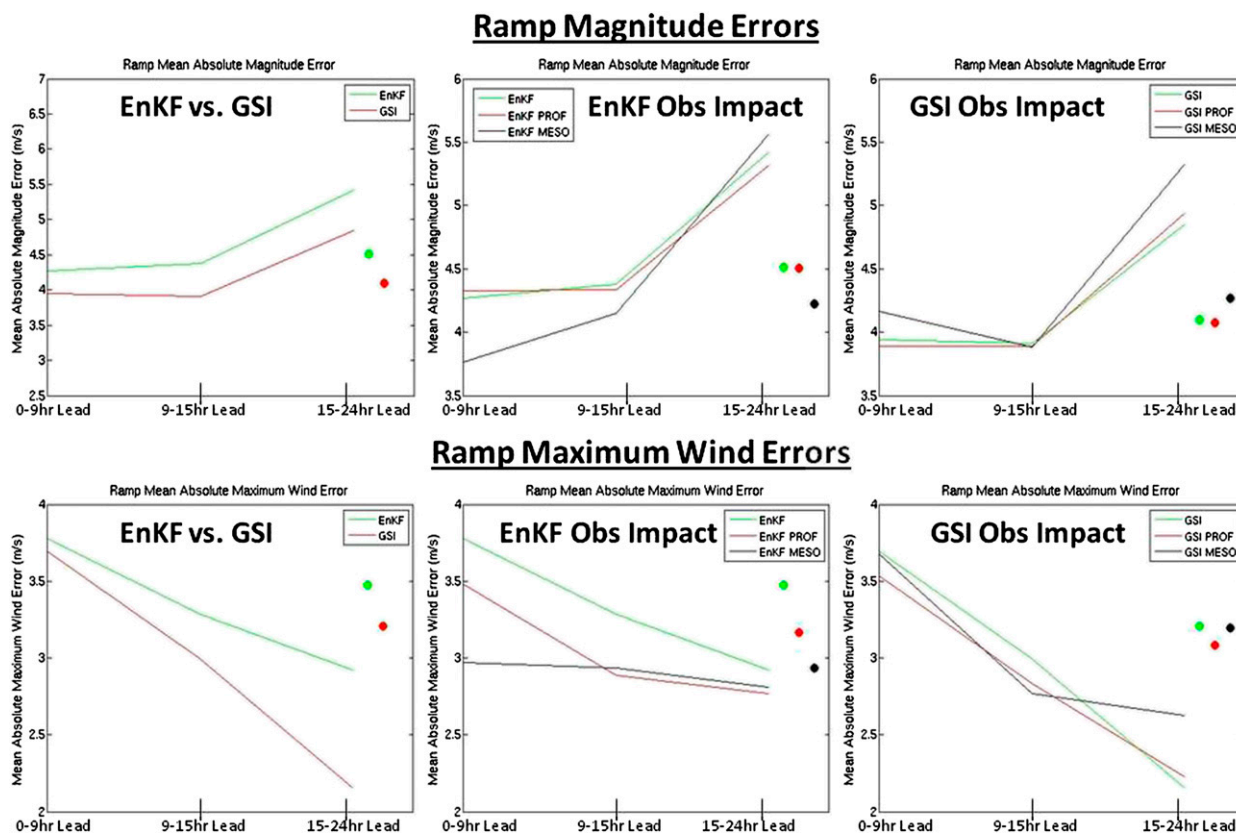


FIG. 11. Ramp magnitude and maximum wind errors averaged over all wind ramp cases for the EnKF and 3DVAR routine and observation impact experiments. The colored circles represent the average error over the three lead-time categories shown (0–9, 9–15, and 15–24 h).

different ramp aspects, and this is also a planned future extension of this work.

4. Summary and conclusions

The first objective of this study was to compare lower-atmospheric wind forecasts from the EnKF, a fully flow-dependent ensemble data assimilation/forecasting system, to that of a deterministic 3DVAR system that uses static covariances. Verification of 24-h forecasts was performed against surface observations over a large area including Texas and Oklahoma over a month-long period. Forecasts from each system for 10 wind ramp cases were also evaluated against 80-m tower data within a relatively small area of existing wind farms in central Texas. For the month-long period, it was found that forecasts were better within the EnKF for nearly the entire forecast period, likely a result of a more realistic analysis achieved through flow-dependent covariance relationships. Both profiler/sodar and mesonet observations had little impact on later forecast hours (6–24 h), likely caused by the influence of like boundary conditions in both experiments. However, mesonet observations

improved early (0–6 h) forecasts in both the EnKF and 3DVAR and thus should be considered beneficial to lower-atmospheric wind prediction. On the contrary, profiler/sodar observations had practically no effect on early forecasts using 3DVAR, and degraded early forecasts in the EnKF, a result that appears linked to the assimilation of wind observations in the presence of a lower-atmospheric wind bias. Mitigating this issue can be done by 1) testing different physics schemes, such as the PBL parameterization, in the hopes of discovering a physics configuration that produces sufficiently small biases, or 2) developing a bias-removal technique that improves the analysis, yet does not degrade forecasts as has been shown in previous work. Until bias is addressed, assimilating lower-atmospheric wind observations such as those from profilers and sodars will likely continue to degrade near-surface wind forecasts within the current modeling configuration.

Wind ramp forecast performance showed practically the opposite behavior compared to that of the month-long experiment. The 3DVAR system outperformed the EnKF for the ramp attributes of onset, duration, magnitude, and maximum winds at early, middle, and late

forecast times within the 24-h window. It was shown that this probably was a result of utilizing the ensemble mean in the presence of substantial nonlinear ensemble evolution, a situation that can cause a poor model representation of sharp features such as wind ramps. In turn, “best member” techniques that select specific ensemble members should be explored to improve the value of the EnKF wind ramp forecasts. The authors plan to examine two such techniques in the near future as an extension of this study: 1) members chosen based on their closeness to the mean for a number of metrics (e.g., the wind ramp attributes, or domainwide sea level pressure), and 2) members selected based on their errors in sensitive regions using a metric defined through ensemble sensitivity analysis. Whether sufficient improvement using these techniques can be obtained to outperform 3DVAR is a key question we aim to answer.

Interestingly, observation impacts on wind ramp forecasts varied. Profiler/sodar and mesonet observations generally improved EnKF forecasts, but generally degraded 3DVAR forecasts. This further reveals the potential of the EnKF to exceed the performance of 3DVAR as observational capabilities are enhanced. It should be noted that the collective results of this study apply to the specific horizontal and vertical resolution used in these experiments, as well as the particular modeling and data assimilation configurations. We also stress that the results found here should be viewed as a benchmark for further investigation into more advanced assimilation systems, such as 4DVAR or hybrid techniques, which have the potential to improve forecasts over that of both the EnKF and 3DVAR.

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